Resampling in Particle Filter Models Produces Online Digging-In Effects

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Introduction: Under surprisal theory [1, 2], linguistic representations only affect processing difficulty through the causal bottleneck of surprisal. Our best estimates of surprisal come from LLMs, which have no explicit representation of structural ambiguity. While LLM surprisal robustly predicts reading times across languages [3], it underpredicts difficulty when structural expectations are violated [4]—suggesting that representations of structural ambiguity are causally implicated in language processing. Particle filter models [5] offer an alternative where structural hypotheses are explicitly represented as a finite set of particles. We prove that their limited parallelism amplifies garden-path effects, and that resampling—a common practice with these models—produces online digging-in effects [6] where disambiguation difficulty increases with ambiguous region length.

Framework: When applied to language processing, particle filter models approximate distributions over structural analyses through a limited set of particles, each representing a candidate parse. With each incoming word, particles are weighted by how well they predict it, then resampled with replacement according to these weights—producing a fresh unweighted sample from the same distribution with some stochastic drift. We isolate the effect of resampling by assuming that lengthening the ambiguous region provides no additional structural or lexical information—a strong but intentional simplification demonstrating that digging-in arises from the processing algorithm alone.

Theoretical Results: We prove two key results regarding resampling's effect on a word's expected surprisal. **(1)** *Expected surprisal increases monotonically with resampling.* This increase is strict unless the word carries no structural information or only a single structure is maintained. **(2)** *Resampling asymptotically increases expected surprisal by the KL divergence from the prior to the posterior over structures.* Thus, resampling introduces a cost associated with ignoring the structural information present in the word. Individuals eventually commit to a single structure, after which observing the word produces no structural belief updates. This cost is greatest when words disambiguate to dispreferred structures, and if the word rules out any prior structures, the cost is infinite.

Intuition for Digging-In Effects: Demonstrating digging-in requires showing expected surprisal increases *faster* for words disambiguating to dispreferred structures. We approximate the per-step increase using two interpretable quantities. **(3)** *A second-order Taylor approximation shows per-step increase is proportional to the expected squared coefficient of variation of word probability across structures.* This is maximized when structures disagree about the word's probability and it is unlikely under dominant structures—precisely the case for words disambiguating to dispreferred structures. **(4)** *A linear-diffusion approximation divides the induced cost (from 2) by convergence times from genetic drift theory [7].* Convergence times are word-independent, so words with greater cost have larger per-step increases. In both approximations, digging-in magnitude is inversely proportional to particle count—fully parallel models predict no digging-in. Simulations (Fig. 3) across particle counts and disambiguation strengths validate both approximations and thus their intuitions.

Garden-Paths: Digging-in effects are a consequence of a more general phenomenon: sample variance increases disambiguation difficulty. Since the initial particle set is also a sample, limited parallelism, in and of itself, produces amplified garden-path effects relative to the true distribution.

Conclusion: Our analysis, under surprisal theory, (a) offers a candidate account of why garden-path difficulty is greater than predicted by LLM estimates [4] and (b) mathematically demonstrates that resampling predicts online digging-in effects—extending empirical work from Levy et al. [8] to cases without parse failure. If empirical evidence fails to support robust online effects, this would argue against models with regular resampling—the set of entertained analyses would need to be selectively updated, keeping low-probability parses around for longer than may seem rational.

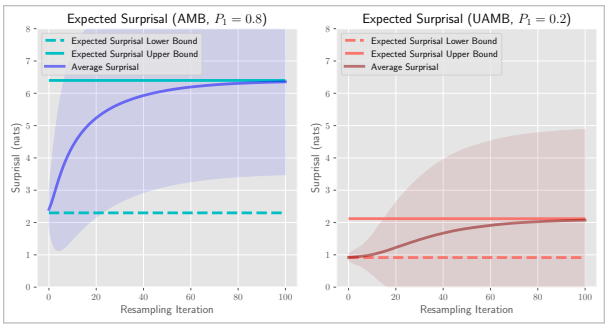
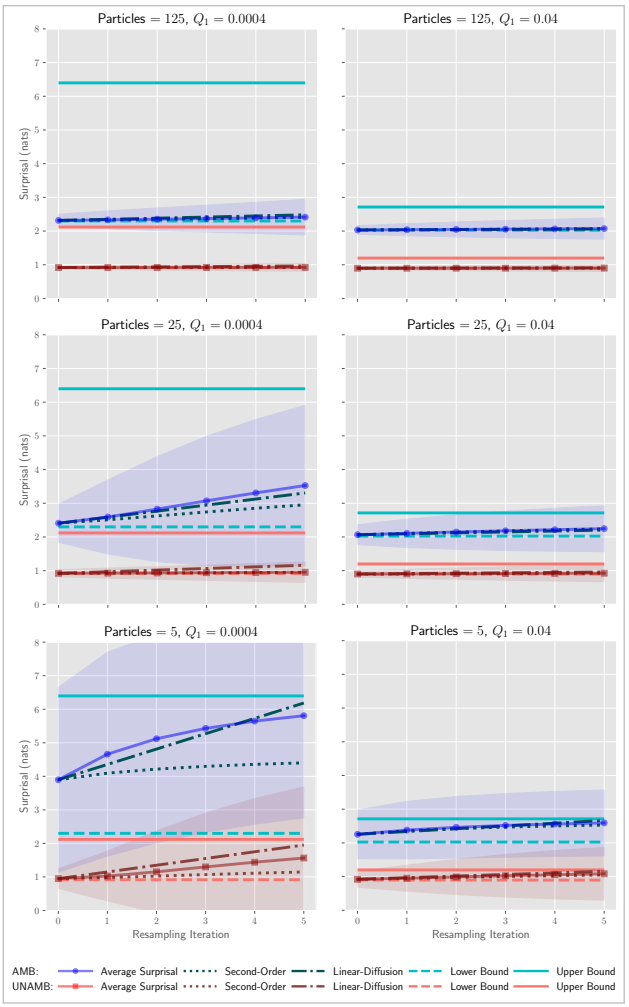
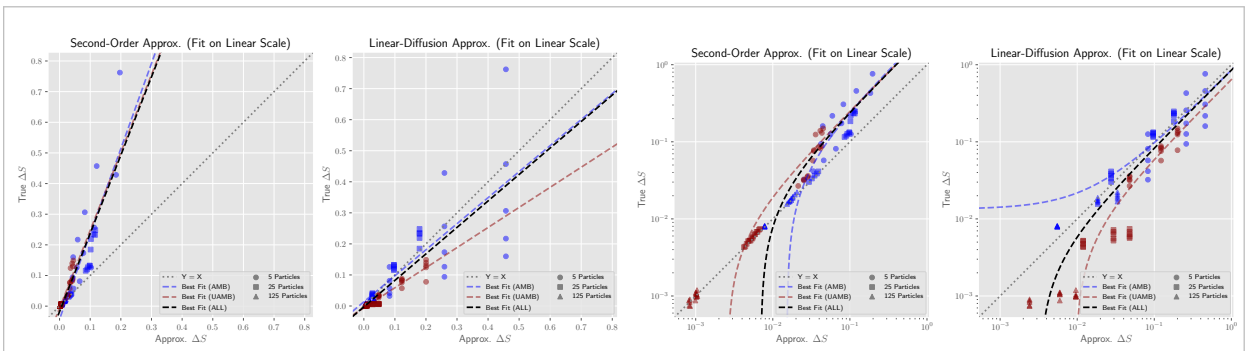


Figure 1 (Above): Expected surprisal over the course of resampling. We assume two structures, T_1 and T_2, which give word w the in-context probability $P(w \mid T_1, C) = 0.004$ and $P(w \mid T_2, C) = 0.5$, such that w disambiguates to T_2. Using 25 particles, we simulate the change in surprisal in an ambiguous context ($P(T_1 \mid C) = 0.8$) and unambiguous context ($P(T_1 \mid C) = 0.2$). Shaded regions indicate ±1 stdev of the sample; error bars would not exceed line width.

Figure 2 (Right): Expected surprisal in early resampling steps alongside the expected surprisal trajectories computed using both the second-order and linear-diffusion approximations. Using the same setting as Fig. 1, we vary disambiguation strength (through $Q_1 = P(w \mid T_1, C)$), and the number of particles. Shaded regions indicate ±1 stdev of the sample; error bars would not exceed line width.

Figure 3 (Below): True per-step surprisal increase by approximated value, using the same data visualized in Fig. 2 (including values for $Q_1 = 0.004$, excluded from Fig. 2 for space). Best-fit lines are computed with data on a linear scale. Linear-diffusion better accounts for large increases, while second-order better accounts for small increases. (Spearman ρ: linear-diffusion = 0.89; second-order = 0.99).





A: Data visualized on linear scale. **B**: Data visualized on log-log scale.

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